B12 Model evaluation

Metric choice, calibration, imbalanced classes and error analysis.

B12-L1 Choose the metric from the decision

Choose metrics and thresholds from the decision. Report prevalence and counts beside every number.

Accuracy at 2.4% prevalence	0.976 for predicting all negatives. Useless as a headline.
Baseline for average precision	The prevalence. Report it beside the score.
Where the threshold comes from	The decision: review capacity or costs. Not the 0.5 default.
Precision and recall	Precision: share of flags that are real. Recall: share of real cases flagged.

B12-L2 Calibration, when the probability itself matters

When the probability is used as a number, check calibration, compare the Brier score with a constant baseline, and recalibrate on held-out data.

Calibrated means	Of cases scored p, about p are positive.
Check calibration	Calibration table or calibration_curve, plus Brier score against the constant prevalence baseline.
class_weight="balanced" and probabilities	Pushes them towards 0.5. Mean p went from 0.024 to 0.42 at 2.4% prevalence.
Recalibrate	CalibratedClassifierCV(model, method="sigmoid", cv=5). Isotonic only with plenty of data.

B12-L3 Imbalanced classes and error analysis by segment

Treat rare positives with a chosen threshold first, then break results down by segment and read the worst errors.

First response to rare positives	Train normally and choose the threshold from capacity or cost.
Resampling rule	Only inside training folds. Never the test set.
Error analysis	Metrics per segment with counts, then read the most confident errors.
Pooled vs segment recall	Pooling weights large segments. Overall 0.7 can hide a segment at 0.2.